

The statistics of soaring flight

Displacement scaling across terrain, circuit types and wing classes

Matteo Di Sante (intern, September 2026 – February 2027)



OFFSITE
2026

EconophysicsLab offsite 2026 | Slide 1

The statistics of soaring flight

The aim is to describe soaring flight as a transport process and to identify empirical constraints for a later stochastic model. Today I cover the archive and its preparation, horizontal mean-square displacement and conditional comparisons through Section 3.2, followed by open questions and next steps.

The cover evokes the offsite setting: Lake Maggiore, Isola dei Pescatori and the Alps. A glide leads into local search and a helical climb in a thermal rising from a sun-warmed mainland slope. Upward filaments indicate airflow; the white helix is the aircraft path, with wind drift. The foreground island includes its bell tower and a red waterfront building without signs. This AI-generated illustration is geographically inspired, not a literal topographic map, measured airflow or a recorded flight.

Visual references: the user-supplied photograph and Comune di Verbania, Isola Superiore or Isola dei Pescatori, including the aerial photograph credited to Robin Ulrich.

The soaring flight cycle

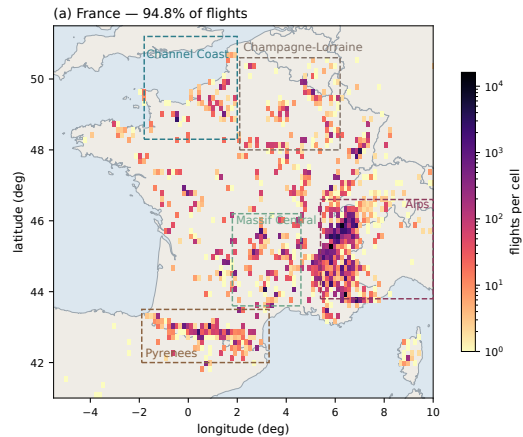
Paragliders and hang gliders have no engine. A cross-country flight repeats three phases. In transition the pilot glides, trading height for horizontal distance. In search the pilot looks for rising air. In climb the pilot circles inside a thermal, a column of air rising where the sun has heated the ground, and regains the height spent. Repeating the cycle, with thermals as the only source of height, is what lets a pilot cover long distances without an engine.

The illustration is the cover image: a glide, a short search and a helical climb in a thermal over a sun-warmed slope. The schematic is illustrative and has no numerical scale. These are behavioural descriptions, not phase labels supplied by the recorder; Vilpellet et al. provide a hidden Markov model that labels each fix as transition, search or climb from binary trajectory features, and slide 14 returns to it. Ground motion includes both air-relative motion and wind. Source: thesis introduction, Sections 1.1–1.3.



Thermals let pilots cover long distances without an engine

A large archive of free flight



FFVL distance competition

186,052

Paraglider tracklogs
1999-2026

6,716

Hang-glider tracklogs
2001-2026

Regional boxes

- Channel Coast
- Champagne-Lorraine
- Massif Central
- Alps
- Pyrenees

IGC trajectories + metadata: time, position, altitude, site, circuit and wing class

The source is the FFVL Coupe Fédérale de Distance, with separate archives for paragliders and hang gliders. The verified downloads contain 186,052 and 6,716 tracklogs. Season names refer to their starting year. The IGC files supply time-stamped coordinates and altitude channels, while the catalogue supplies flight date, declared take-off site, route class and wing class. The map shows the retained population, not the raw downloads; both disciplines are pooled. About 94.8 percent of retained flights are in metropolitan France, with a strong Alpine concentration. The coloured boxes are the longitude–latitude windows used for regional comparisons, listed on the right in the same colours: Channel Coast, Champagne-Lorraine, Massif Central, Alps and Pyrenees. Additional flights occur in La Réunion and elsewhere. This is a submitted cross-country archive with uneven geographic coverage, not a random sample of all free flight. Sources: thesis Chapter 2; generated/stats.tex, pipeline_census.tex and prelim_map.pdf.

How do we study the statistics of this motion?

Mean-square displacement (MSD)

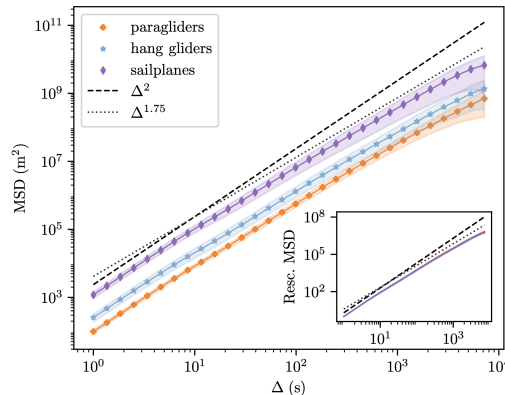
- 1 Along each flight f , of duration T_f

$$m_f(\tau) = \frac{1}{T_f - \tau} \int_0^{T_f - \tau} |x_f(t + \tau) - x_f(t)|^2 dt$$

- 2 Over the N flights: $M_2(\tau) = \frac{1}{N} \sum_f m_f(\tau)$

$$M_2(\tau) \propto \tau^{2H}: H = \frac{1}{2} \text{ diffusive, } H = 1 \text{ ballistic}$$

Vilpellet et al.: French flights, 2016–2021



$M_2(\tau) \propto \tau^{2H}$ with $H \approx 0.88$: superdiffusive over three decades, 10^1 – 10^4 s

Vilpellet, Darmon and Benzaquen analysed publicly available French tracks from 2016 to 2021: CFD–FFVL for paragliders and hang gliders, NetCoupe for sailplanes. After requiring at least 1 Hz sampling, one hour of flight, no discontinuities and a 15 km horizontal range, 49,939 paraglider, 780 hang-glider and 9,203 sailplane trajectories remain. The paper states no terrain or regional selection. Their Figure 1 shows the horizontal MSD. The paper writes it as an expectation, $\delta^2(\Delta) = \mathbb{E}[(x_{t+\Delta} - x_t)^2]$, under a stationarity assumption; it is estimated as on the left: a time average along each flight, then an average over flights. The slide uses the notation of the rest of the talk: τ for the lag, m_f per flight and M_2 over flights. Slide 22 applies the same estimator within the retained segments of each flight, and the exponent H of the bottom line is the one compared there. Below about 10 s the growth is ballistic, $\delta^2 \sim \Delta^2$. At longer lags it is sub-ballistic but faster than diffusion, $\delta^2 \sim \Delta^{2H}$ with $\frac{1}{2} < H \approx 0.88 < 1$. The three curves differ mainly by a vertical shift set by their characteristic speeds, and the speed-rescaled curves in the inset nearly collapse. They attribute the exponent to the intermittent, two-dimensional and directionally correlated transition–search–climb cycle. They also provide a hidden Markov model that labels each fix as transition, search or climb from binary trajectory features; slide 14 returns to it.

The challenge is to explain whole-flight transport while respecting the energy budget of the previous slide. The present measurements are the first layer of that programme. An MSD exponent alone cannot distinguish all the candidate mechanisms: persistent motion, changes of behaviour and heterogeneous

Preprocessing

Clean the recorded flight

Selection and coordinates

Uniform time grid

Savitzky–Golay smoothing

Retained (of downloads)

156,406

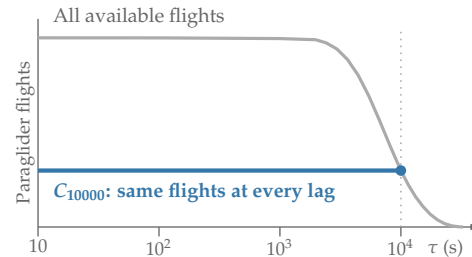
paragliders, 84%

6,094

hang gliders, 91%

Fixed cohort C_{10000}

Segments of at least 12,500 s (≈ 3.5 h)



In C_{10000} (of retained)

46,273

paragliders, 30%

2,326

hang gliders, 38%

From raw tracklogs to the fixed cohort C_{10000}

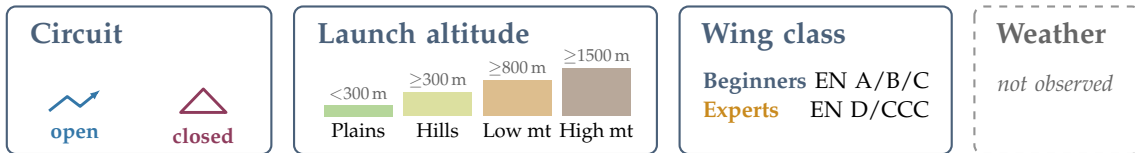
Details are in the appendix: each step heading, and the cohort heading, links to its slide; each appendix slide links back here.

Seven stages turn the downloads into cleaned flights: impossible fixes and ground phases are removed; flights need at least 40 minutes, 20 km and 600 m of altitude range; positions become local east–north metres; a uniform grid at native cadence is built, and long gaps split a flight into segments; a moving cubic Savitzky–Golay fit gives position, velocity and acceleration. 156,406 paraglider and 6,094 hang-glider flights remain.

A segment enters at lag τ only if $\tau \leq 0.8 T_s$. Without a fixed cohort (grey), the paraglider count falls from 155,085 at 10 s to 46,273 at 10,000 s, so any lag-dependent statistic would mix dynamics with a changing set of flights. Applying the rule once, at $\tau_{\max} = 10,000$ s, gives $T_s \geq 12,500$ s: 46,273 paragliders and 2,326 hang gliders, the same at every lag; the two curves meet at 10,000 s by construction. The cost: on matched lag ranges these long flights have H higher by 0.016–0.018, about 2 percent. Sources: thesis Sections 2.7–2.8 and 3.1; pipeline_census.tex, ch3_transport_report.json.

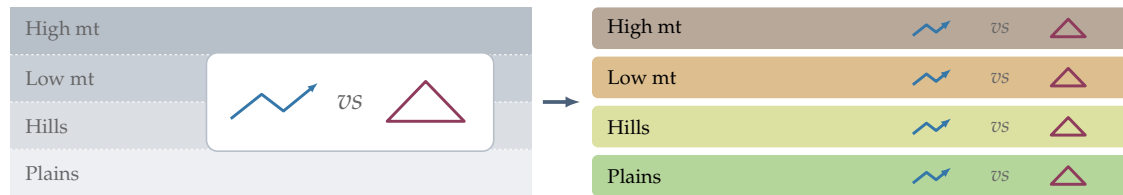
Which flight characteristics shape MSD and H ?

6



Marginal across all altitude classes

Stratified within each altitude class



First compare groups overall, then within the same altitude class.

Altitude class set by z_0 , the GNSS altitude of the first fix after trimming.

Which flight characteristics shape MSD and H ?

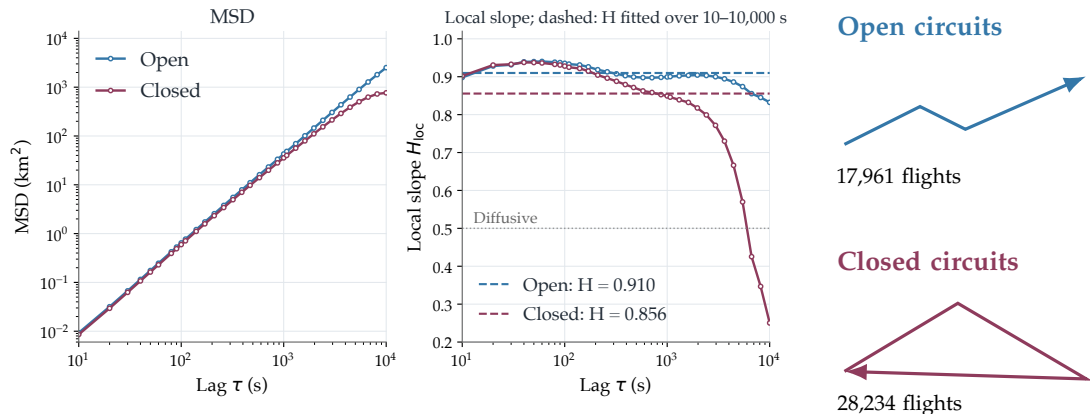
Each flight is assigned to an altitude class by z_0 , the GNSS altitude of the first fix of the trimmed track, that is, the start of free flight rather than the first fix of the raw log. Thresholds are 300, 800 and 1500 metres. Weather also matters, but we have no data with sufficient spatial and temporal resolution.

We examine circuit type, launch-altitude class and wing class. Launch altitude describes the starting environment, rather than the full terrain traversed. Wing class reflects equipment performance as well as a proxy for pilot experience.

A marginal circuit comparison pools altitude classes: open and closed flights can have different altitude compositions. Stratification compares open and closed flights within Plains, then within Hills, Low mountains and High mountains. The altitude class is fixed, while equipment, weather and the precise altitude within that class can still differ. The comparison is conditional on altitude class and still marginal over the remaining characteristics. Stratification does not require all flight conditions to be identical.

The next figures show pooled circuit and altitude contrasts, then examine circuit composition and its contribution to the pooled MSD. We then compare circuits within altitude classes: does the open-closed contrast persist, and does its size change? It persists but narrows in mountains, and the altitude ordering differs between open and closed circuits. Later comparisons examine regions and wing classes within selected altitude categories.

Open vs closed: how do MSD and H differ?



$H_{\text{open}} - H_{\text{closed}} = 0.054 [0.053, 0.056] > 0$ (90% paired interval)
Slower closed-route MSD growth is consistent with turns and returns.

78 unclassified flights excluded. Nominal 90% cluster-bootstrap CIs; bands likely too narrow (Appendix).

The catalogue describes the intended circuit, not a guarantee that the trimmed trajectory returns to its starting point. Closed routes include triangles and out-and-return flights. Pooling launch altitudes gives 0.910 [0.909, 0.911] for open routes and 0.856 [0.855, 0.856] for closed routes. Their paired difference is positive under the adopted bootstrap.

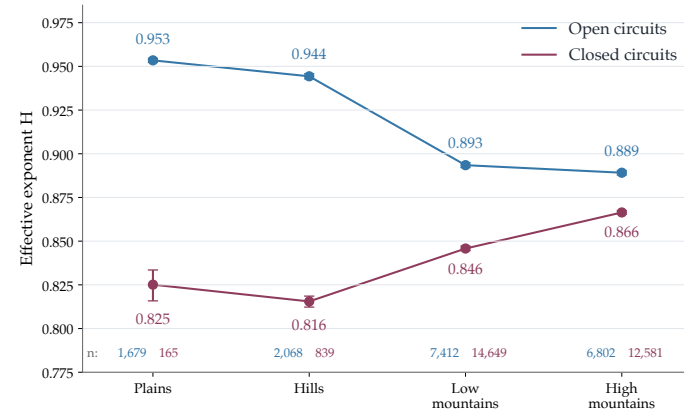
The right panel is the local slope H_{loc} of the same two curves, with the estimator of slide 5 applied to the 33 archived lags: point estimates only, no band. Dashed lines are the fitted H values. Up to about 100 s the two local slopes agree within 0.01. Beyond, closed circuits fall, to 0.25 at 10,000 s, while open circuits stay above 0.83. The fitted gap therefore comes from the long lags.

We fix the lag and average squared displacements over starting times along each segment. Suppose an ideal pilot reverses direction at minute 60, with the same speed on both legs. A two-minute interval from minute 59 to 61 spans the reversal and ends at its starting position; minute 20 to 22 covers straight flight. Thus even a short lag can capture a turn late in the flight. With equal leg times, $T/2$ and $T/3$ locate the first major turn along out-and-return and triangular routes. They do not define a minimum affected lag. Ten thousand seconds is the analysis limit; flight and leg durations vary. Testing the triangle hypothesis requires separating route types and examining actual leg durations.

The schematics illustrate geometry only. This marginal comparison pools launch environments and equipment, so its magnitude cannot rank the causal importance of route choice against terrain. Sources:

Does the open–closed H gap change with altitude?

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Open: higher H in lowlands

- Lowlands: no deviations for lift
- Mountains: deviations to known lift

Closed: higher H in mountains

- Lowlands: turn back any time
- Mountains: way home follows ridges

Open > closed everywhere: gap 0.13 in lowlands, 0.02–0.05 in mountains.
Mountain terrain constrains both circuits, so the gap narrows.

Nominal 90% bootstrap CIs; clusters sampled with replacement. Clustering remains too fine; bands likely too narrow.

Does the open–closed H gap change with altitude?

Read this figure in two directions. Comparing circuits within each altitude class gives open H greater than closed H in all four classes. The paired differences are 0.128 [0.120, 0.138] in Plains, 0.129 [0.125, 0.132] in Hills, 0.048 [0.046, 0.049] in Low mountains and 0.023 [0.021, 0.024] in High mountains. All four nominal 90 percent intervals are above zero. The circuit ordering survives altitude stratification, so different altitude composition alone cannot explain it. This is a consistent conditional association; its size still depends on altitude.

Comparing altitudes within each circuit reveals the reversal. Lowland-minus-mountain H contrasts are positive for open routes (0.051–0.064) and negative for closed routes (magnitudes 0.021–0.051). All eight paired intervals exclude zero. There is no general mountain penalty in H .

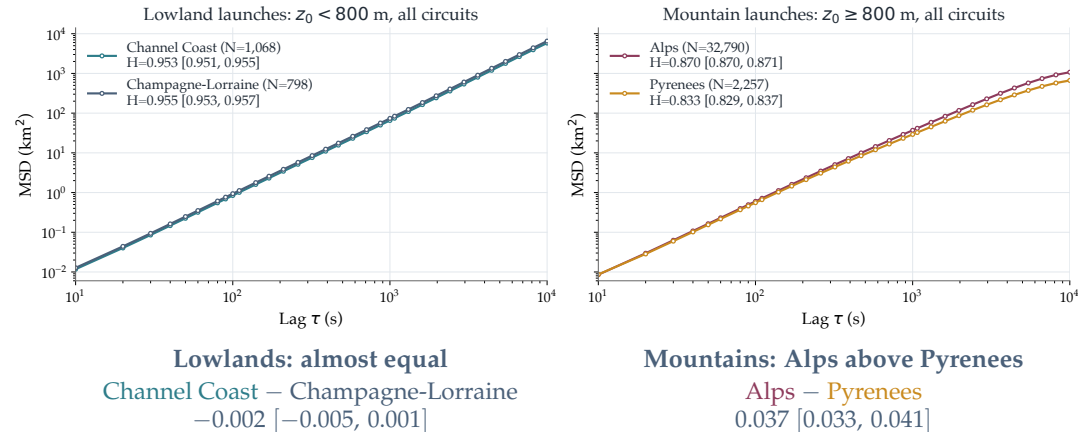
The two findings combine into a narrower circuit gap in mountains, as quantified by the four paired estimates above. This is the altitude-by-circuit interaction of the factorial appendix: a lowland gap minus a mountain gap equals the open lowland-minus-mountain shift minus the closed one, a positive minus a negative number, so it is positive. Our interpretation is that orography constrains how pilots carry out either task. In mountains, usable thermals are localised on sun-facing slopes and along ridges, so each transition must be planned towards places where lift is known to form. In plains, lift use is spread more evenly in space and in height, with fewer clusters: a pilot meets thermals here and there along the way. Open circuits are therefore somewhat less open in mountains. Glides bend

How does H differ by region?

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How does H differ by region?



H separates the two mountain ranges more clearly than the two plains.

Nominal 90% bootstrap CIs; clusters sampled with replacement. Clustering remains too fine; bands likely too narrow.

Open and closed circuits both enter this slide, as in thesis Section 3.2.3. The regional selections hold broad launch altitude fixed: below 800 m in Channel Coast and Champagne-Lorraine, at least 800 m in the Alps and Pyrenees. Legends give the selected flight counts: 1,068 on the Channel Coast, 798 in Champagne-Lorraine, 32,790 in the Alps and 2,257 in the Pyrenees. Within each pair the circuit mix is similar: open routes are 1,020 and 741 of the lowland flights (about 95 percent), 10,533 and 698 of the mountain flights (about a third).

The lowland difference is -0.002 [$-0.005, 0.001$]: practically zero, and its nominal interval includes zero. H is 0.870 [0.870, 0.871] in the Alps against 0.833 [0.829, 0.837] in the Pyrenees, a paired difference of 0.037 [0.033, 0.041], about twenty times the lowland one. These bands are likely too narrow, so neither interval should be read as a significance test. The claim is the comparison: H separates the two ranges more clearly than the two plains. Restricted to open routes, the mountain difference shrinks to 0.011 [0.006, 0.016] and the lowland one is -0.003 [$-0.006, -0.001$]: part of the Alps–Pyrenees gap comes from closed routes, but the ordering survives.

Same cohort, estimator, 10–10,000 s fit and archived paired site–day draws as the thesis; the open-route check is `measure_open_circuits.py`.

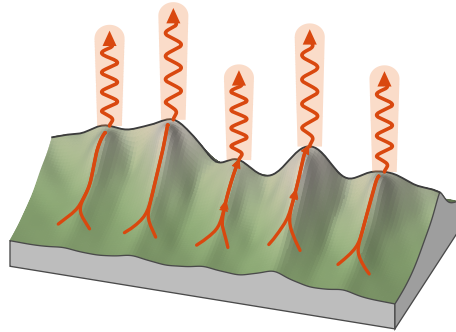
A possible explanation is the constraint set by the terrain. In mountains, thermals form on sun-facing slopes and spurs and release along the crests, so usable lift follows the orographic profile of each

Why does H separate the ranges but not the plains?

10

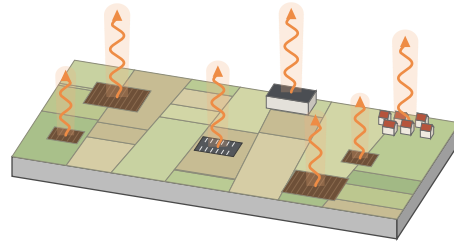
Mountains: lift follows the slopes

Strong thermals climb sunny spurs to the crest.



Lowlands: scattered triggers

Fields, asphalt, roofs: random spots, weaker lift.



Mountain lift traces each range's relief; lowland lift is scattered at random.

So the Alps and Pyrenees can differ in H , while two plains with similar trigger statistics cannot.

Orange: rising air. Illustrative schematic, not measurements. Sources: FAA, Ch. 9; SSA, Lift Sources.

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Why does H separate the ranges but not the plains?

This schematic illustrates the explanation proposed on the previous slide. In mountains, sun-facing slopes heat first. Warm air flows up the spurs and leaves the terrain along the crest, so thermals are strong and sit where the relief puts them. Their pattern traces the orographic profile of the range: another range, with other ridges and spurs, gives another pattern, and pilots must fly between those fixed places. This is one way the Alps and the Pyrenees can differ in H .

In the lowlands, thermals are weaker and start from heated surfaces of many kinds: ploughed fields, a car park, a warehouse roof, a village. These triggers are scattered with no geometric pattern, and a heated field need not release a thermal at a given moment. If two plains share the same trigger statistics, pilots meet lift in statistically equivalent places, and H is the same.

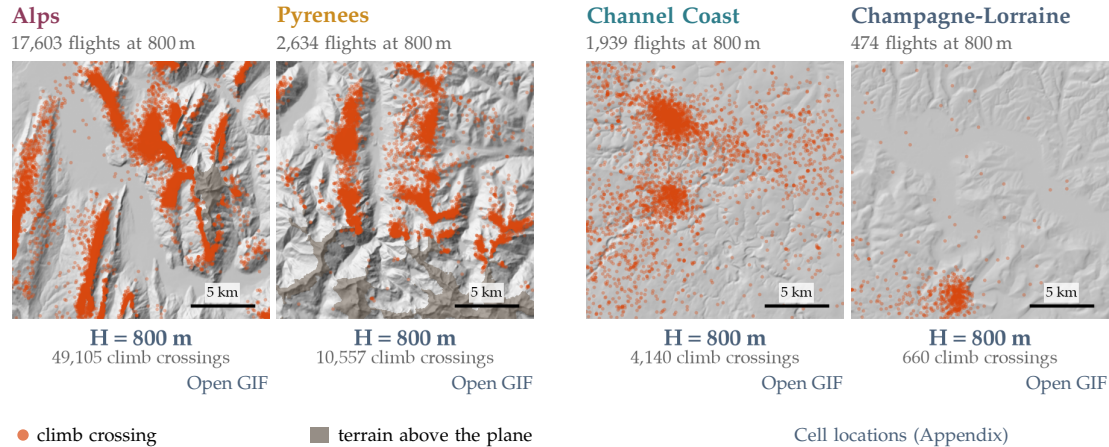
The drawing carries no wind and no data: it is a conceptual picture. Testing it needs the spatial statistics of climbs in each region, as in the appendix maps, compared across ranges and across plains. Sources: FAA Glider Flying Handbook, Chapter 9; SSA, Lift Sources; thesis Section 3.2.3.

Mountain lift follows ridges, lowland lift spreads aloft

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Mountain lift follows ridges, lowland lift spreads aloft



Higher up, a random location may offer easier access to thermals in lowlands.

This may explain similar Hurst exponents across lowlands and differences between mountain ranges.

Observed climbs, both disciplines, all dates. Plane $H = 200\text{--}1,400$ m above mean terrain. Terrain: © IGN.

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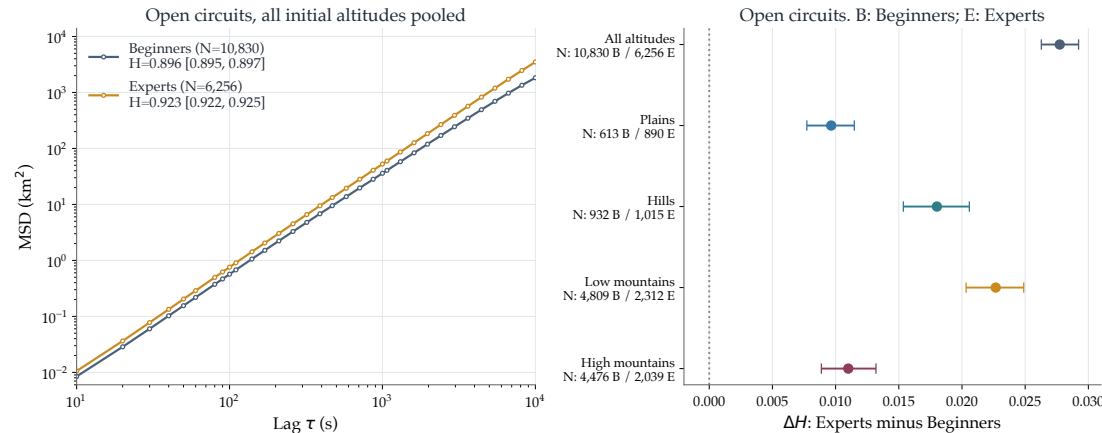
Each map shows the busiest 20 by 20 km cell in its region, selected by distinct flights across all dates and both disciplines. Orange dots mark own-HMM climb crossings; brown marks terrain above the horizontal plane. Here H is height above the cell's mean IGN terrain, distinct from the Hurst exponent. Mean terrain: 879, 1,796, 172 and 218 m ASL, respectively. The appendix gives exact locations.

At 800 m, the four maps contain 49,105, 10,557, 4,140 and 660 crossings. Contributing flights and crossing counts vary with height. A flight may cross repeatedly. Mountain crossings trace ridges; lowland crossings spread from launch clusters into broader clouds aloft. This suggests easier access to thermals at a random location in lowlands than in mountains. It does not imply increasing crossing counts.

Possible link to Hurst: mountain routes follow each range's lift geometry, while similar lowland lift statistics could produce similar exponents. These selected cells illustrate a hypothesis; regional Hurst estimates use larger populations. Traffic and weather affect coverage, so these maps do not measure encounter probabilities. At 800 m, Champagne-Lorraine is as compact as the mountain cells: shape and its height dependence motivate the interpretation.

Possible collaboration: Dr. César Hernández Aguayo. Sources: plane-cell measurement reports and rendering scripts.

Open circuits: expert–beginner H gap by altitude



Vilpellet et al. also found $H_{\text{expert}} > H_{\text{beginner}}$ with experts spending less time searching for thermals.

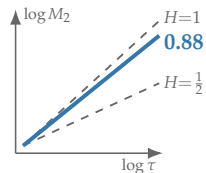
Wing class is a proxy. Nominal 90% cluster-bootstrap CIs (Appendix). Vilpellet et al.: arXiv:2601.01293, Figs. 5a, 6d.

Only open circuits enter this slide, so circuit type cannot drive the contrast; the thesis, Section 3.2.4, pools all circuits. Beginners fly 10,830 of their 27,609 flights on open circuits and Experts 6,256 of 16,497. These are the operational groups defined on slide 6. Individual experience is not observed: an experienced pilot can fly a lower-class wing, and the same pilot can contribute to different groups on different flights.

On open circuits the pooled exponent difference is 0.028 [0.026, 0.029], against 0.025 [0.024, 0.026] with all circuits. Within altitude classes the differences are 0.010 [0.008, 0.011] in Plains, 0.018 [0.015, 0.021] in Hills, 0.023 [0.020, 0.025] in Low mountains and 0.011 [0.009, 0.013] in High mountains, all with paired intervals excluding zero. The Low mountains gap exceeds the other three in the paired interaction tests: -0.013 [-0.016 , -0.010] against Plains, -0.005 [-0.008 , -0.001] against Hills and 0.012 [0.009, 0.015] against High mountains, each written as the first class's gap minus the second's. Plains and High mountains are indistinguishable, -0.001 [-0.004 , 0.001].

Vilpellet et al. found the same ordering with three EN groups: H rises from about 0.85 for beginner wings (EN A and B) to about 0.92 for competition wings (EN D and CCC), our Experts. The fraction of flight time spent searching decreases systematically with wing class (their Figs. 6d and 7), and the search-phase exponent rises from about 0.26 to 0.35. They conclude that skill acts mainly during search: it raises the thermal detection rate and so shortens the search.

Scaling



Strongly superdiffusive

- Paragliders $H = 0.88$
- Hang gliders $H = 0.85$

Terrain



Mountains narrow the gap

- Open-route H falls
- Closed-route H rises

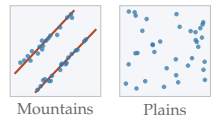
Route and wing



Open routes spread faster

Experts spread faster

Lift geography



Mountains: ridge climbs

Plains: climbs spread out

Strongly superdiffusive everywhere; route and terrain set how strongly.

Take-home messages

Four findings, one per cell; the band gives the synthesis. What we are still working on follows on the next slide.

Scaling. Over 10 to 10,000 s the fixed C10000 cohort gives $H = 0.882$ for paragliders and 0.852 for hang gliders, between diffusion at one half and ballistic motion at one. The same flights enter at every lag, so the exponent is not produced by long lags keeping only long flights. The pictogram is schematic: the reference lines are the diffusive and ballistic slopes, the blue line has the fitted paraglider slope. H summarises the fit range: the local slope stays near 0.85–0.94 up to about 2,000 s and then declines, to about 0.64 for paragliders and 0.41 for hang gliders at 10,000 s (slide 22, right panel).

Route and wing. Pooled over altitude classes, open circuits reach 0.910 [0.909, 0.911] and closed circuits 0.856 [0.855, 0.856], a paired difference of 0.054; open exceeds closed within every altitude class. On open circuits, Experts exceed Beginners in every class too, by 0.028 pooled and 0.010–0.023 within classes. The canopies are schematic: the longer, thinner one stands for the higher aspect ratio of EN D and CCC wings.

Terrain. The miniature plots the eight circuit-by-altitude exponents of slide 8. Open routes have higher H in the lowlands, closed routes in the mountains, so there is no general mountain penalty. The open–closed gap is 0.128 [0.120, 0.138] in Plains and 0.023 [0.021, 0.024] in High mountains. Because Plains are 91 percent open circuits and the mountain classes about one third, a ranking by altitude

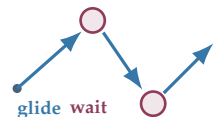
Work in progress: resolving phases and directions

Segmentation



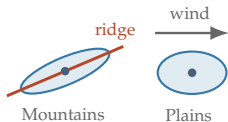
Revise Vilpellet's HMM

Stochastic model



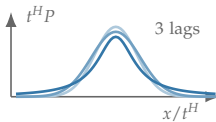
Which model fits best?

Anisotropy



Ridge axis or wind

Distribution scaling



Do distributions collapse?

Solo vs group flight



Does group flight change H ?

The current Vilpellet segmentation is useful and already supplies the thermal-plane examples. The source study retained trajectories at a minimum frequency of one hertz; its appendix uses binary trajectory features. That domain does not validate transfer to our slower loggers. Visual inspection also suggests some climbing or lifting stretches are missed. These are diagnostics, not a completed accuracy benchmark. The proposed replacement remains an HMM with discrete glide, search and climb states, but continuous-valued observations such as vertical speed and turning measures. Features must be defined in physical time and state persistence must account for sampling interval. Merely replacing binary emissions with Gaussians does not solve cadence dependence. Validate on reviewed trajectories and deliberately downsampled high-frequency tracks.

The top-right cell states the goal: compare a set of candidate stochastic models and find which one best describes the empirical observations. One candidate has two phases: glides generate displacement, and each contiguous search and thermalling block becomes one waiting episode. Estimate glide lengths, durations, directions and turning-angle memory, and total waiting times without assuming independent search and climb durations. Position is held fixed during waiting, which neglects local search motion and thermal drift. Monte Carlo simulations with the observed sampling then test MSD, displacement distributions, anisotropy and temporal correlations. Details are in the appendix.

For anisotropy, estimate the displacement tensor at each lag and compare its main axis with local ridge

Thank you

Thank the audience and open the discussion. Backup material follows: the preprocessing stages and the choice of the fixed cohort, then velocity-memory definitions and the bootstrap method.

Thank you for your attention!

Appendix: clean the recorded flight

↔ slide 5 16

1 GNSS altitude

GNSS height → Trajectory $z(t)$

Barometer → Cleaning checks

GNSS: absolute height without weather drift.
Same high-frequency noise as the barometer.

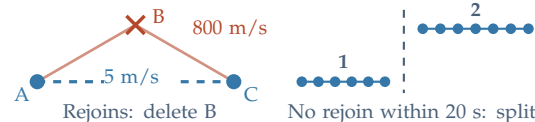
3 Airborne interval



Trim before take-off and after landing.
Stops are not labeled: we infer them from position and altitude.

2 Fix-level cleaning

One fix = (t, φ, λ, z) : time, latitude, longitude, height
Delete an unreachable fix only if the track rejoins without it; otherwise split the flight.



● Retained flight ■ Ground removed

Appendix: clean the recorded flight

GNSS height is the chosen output channel, used throughout. Its absolute reference needs no pressure or temperature calibration, whereas pressure altitude shifts with the weather. In supported paired records, high-frequency median spectra are comparable to barometry. This does not make GNSS uniformly more accurate: logged vertical datums (ellipsoid or geoid) remain unaudited. Barometry corroborates cleaning decisions without replacing the output altitude.

A fix records time, position and altitude at one instant. The rule is conservative: a fix is deleted only when the flight without it is physically consistent. In the left example, A is at 0 m at 0 s, B at 800 m at 1 s, and C at 10 m at 2 s. Reaching B from A requires an impossible 800 m/s. Removing B leaves 10 m in 2 s: 5 m/s. Deletion therefore passes both the impossibility and rejoin tests. The dashed chord tests reachability; it is not interpolation. Contiguous blocks are handled the same way. In the right example the track jumps and stays displaced: no later fix is reachable from the last accepted one within 20 seconds, the Hampel half-window. Unlike the single spike on the left, the displacement persists past the window, so it cannot be an isolated error. The pipeline then keeps both branches, segments 1 and 2, and splits the flight at the jump without deleting any fix, because it cannot tell which branch holds the true positions. A Hampel flag alone never deletes a fix; timestamp, frozen-lock and altitude defects have separate rules.

The timeline distinguishes take-off, intermediate landing, relaunch and final landing. Grey ground

4 Flight selection

Minimum flight thresholds

Time ≥ 40 min

1.13% excluded ($N = 191,108$)

Path ≥ 20 km

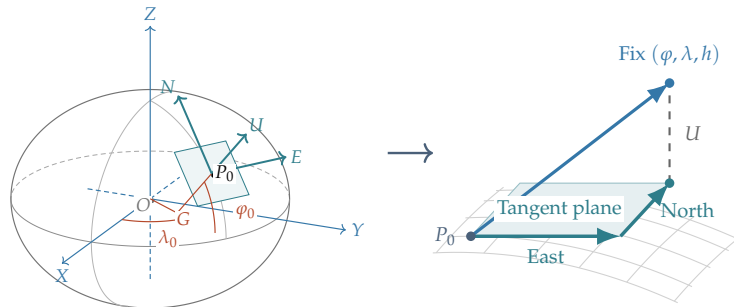
0.09% excluded ($N = 188,942$)

Altitude range ≥ 600 m

2.68% excluded ($N = 188,768$)

5 Coordinates

Latitude, longitude, height $\rightarrow$ Earth-centred (X, Y, Z)
 $\rightarrow$ local east-north distances in metres



Earth model: WGS84 ellipsoid (GPS reference)

P_0 : first trimmed fix; fixed for the flight. $z = h_{\text{GNSS}}$, not U .

Appendix: selection and local coordinates

The minima are 40 minutes, 20 kilometres and 600 metres of altitude range. Elapsed span is capped at 16 hours, with additional integrity and reach checks. Each bar pools both disciplines and represents the logs reaching that sequential check. Duration rejects 2,163 of 191,108 (1.13 percent), path rejects 174 of 188,942 (0.09 percent), and altitude range rejects 5,054 of 188,768 (2.68 percent). Three overlong logs leave before the path check. Blue marks the share passing that check. These are cascade rates, rather than independent tests on the raw archive.

Latitude, longitude and logged GNSS height give an Earth-centred Earth-fixed position. The conversion uses the WGS84 ellipsoid rather than a spherical Earth, because GPS positions are expressed in that reference: semi-major axis 6,378,137 m, flattening $1/298.257223563$. Subtract the first trimmed fix, P_0 , then rotate into local east, north and up. The global schematic preserves the ellipsoid normal with exaggerated flattening. The close-up projects a later fix onto the fixed tangent plane. Keep the same origin and orientation for every segment of that flight. P_0 is near, but need not coincide with, true take-off.

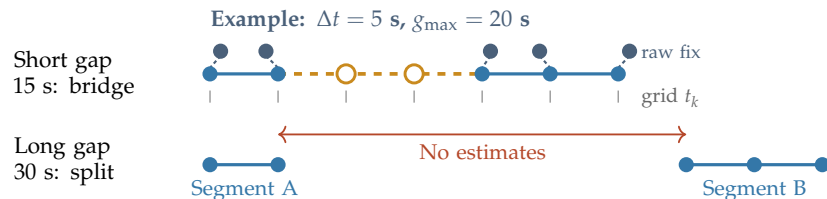
The output vertical coordinate remains logged GNSS height, without subtracting origin height. Rotated Up contains curvature and is not used for vertical dynamics. Final retained counts come after stage 7, on slide 19. Sources: thesis Sections 2.7–2.8, local ENU figure and pipeline_census.tex.

6a Gaps: bridge or split

Gap g between consecutive fixes:

- $g \leq g_{\max}$: bridge it.
- $g > g_{\max}$: split into segments.

$$g_{\max} = \min(10\Delta t, \max(20\text{ s}, 2\Delta t))$$



6b Time grid: jitter removal

Per segment: grid $t_k = t_0 + k\Delta t$, native Δt .

- Interpolate E , N , z at every t_k .
- No fix within $\Delta t/2$ of t_k : flag it as reconstructed.

- Raw fix (jittered)
- Grid point
- Flagged as reconstructed



Keep a segment only if it lasts $\geq 90\text{ s}$ and $\leq 10\%$ of its grid is reconstructed.
No segment kept: flight dropped. 20,798 paragliders, 193 hang gliders: the largest loss.

Appendix: uniform time grid

Stage 6 has two parts. 6a: a gap between consecutive fixes up to $g_{\max}$ is bridged; a longer one splits the flight into segments with no estimates between them. $g_{\max}$ is the smaller of 10 native intervals and 20 seconds, with a floor of two intervals that allows one missed fix; the floor matters only above a 10-second cadence. At a 5-second cadence the cap is 20 seconds. Mandatory cleaning boundaries always split. The same cap applies between finite altitudes: long altitude holes split the full trajectory.

6b: each segment gets an exact uniform grid, t_0 plus k times the native interval (the median recording interval). E and N are interpolated with PCHIP and z linearly at every grid time, not only inside gaps. A fix landing on the grid is reproduced exactly, so an already uniform series passes unchanged; jitter is removed while the flight keeps its own cadence. The half-step rule only sets the flag: a grid time with no fix within half a step is flagged as reconstructed. In the example the 15-second gap leaves two grid times with no nearby fix: orange dashes and hollow markers. Each blue grid point has a raw fix within half a step, joined by a dotted line. Reconstructed grid fixes are 0.58 percent for paragliders and 0.30 percent for hang gliders. Each resulting segment must last at least 90 seconds and have at most 10 percent reconstructed grid times; a failing segment is dropped alone. A flight with no surviving segment is dropped: 20,798 paraglider and 193 hang-glider flights, the largest single loss in the pipeline. Retained counts follow on the next slide, after stage 7. Sources: thesis uniform-sampling section; resample.py.

7 Savitzky–Golay fit

Fit a cubic to E , N , z in a moving window, within each segment.

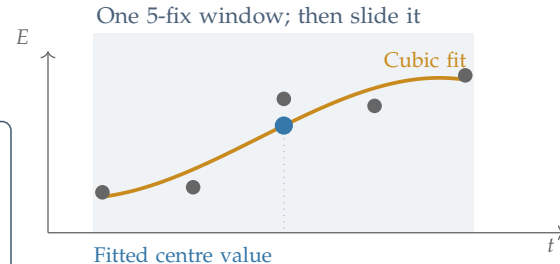
Window: odd, ≥ 5 fixes.

Velocity and acceleration: derivatives of the fitted cubic

$$v_i = P'_i(0), \quad a_i = P''_i(0), \quad u = t - t_i$$

Noise std, for an independent error σ per fix:

	velocity	acceleration
Fitted cubic	$0.95 \sigma / \Delta t$	$0.53 \sigma / \Delta t^2$
Fix-to-fix differences	$1.41 \sigma / \Delta t$	$2.45 \sigma / \Delta t^2$



Retained flights after all seven stages (share of downloaded logs)

156,406 paragliders 84.1%

6,094 hang gliders 90.7%

Schematic; the cubic fit is exact for the synthetic fixes shown.

Appendix: smoothing and derivatives

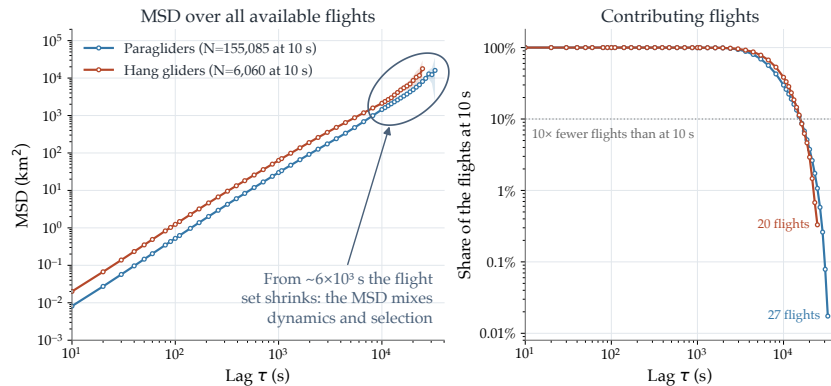
Stage 7 fits a cubic independently to E , N and z in a moving window. With physical time offset $u = t - t_i$, the fitted value and successive polynomial derivatives at zero give position, velocity and acceleration. Derivatives through order three are available from a cubic; the pipeline exports the first two. No subsequent finite differences are taken: those would amplify high-frequency position noise. One least-squares fit over five fixes gives position, velocity and acceleration together. The table gives the noise standard deviation for an independent position error sigma per fix: the norm of the fixed weights each estimate applies to the fixes. Differencing consecutive fixes gives root 2, 1.41 sigma over delta t, for velocity and root 6, 2.45 sigma over delta t squared, for acceleration; the fitted cubic gives 0.95 and 0.53. The gain is modest on velocity and about 4.6 times on acceleration. With five fixes the velocity weights coincide with the classic five-point central difference; its advantage over a three-point central difference, 0.71 sigma over delta t, is accuracy rather than noise. Halving delta t doubles velocity noise and quadruples acceleration noise. The grey synthetic fixes and orange curve show an exact least-squares fit; the blue dot marks its centre value; the derivatives are those of the orange cubic at that time.

The 5-second configured timescale is converted to an odd window with at least five fixes. Five fixes span 4 seconds at 1-second cadence but 20 seconds at 5-second cadence. Smoothing can therefore affect short-lag statistics differently across cadences. Windows never cross a split; one-sided edge estimates and reconstructed support remain flagged. Retained counts cover the full pipeline: 156.406

Appendix: which flights contribute at each lag?

↔ slide 5 20

All available flights: a flight enters at lag τ if one of its segments has $\tau \leq 0.8 T_s$ ($\geq 1/5$ of it left as origins for m_f).



At long lags only long flights remain, and their number falls steeply.

Any lag-dependent statistic then mixes dynamics with a changing set of flights.

Desiderata

- 1 The same flights at every lag
- 2 The longest possible lag range
- 3 $N \approx$ of the same order as at 10 s

Appendix: which flights contribute at each lag?

Before fixing a population, look at what happens without one. At each lag we average over every flight that can contribute: a flight enters at lag τ if at least one of its retained segments satisfies $\tau \leq 0.8 T_s$. At least a fifth of that segment then remains available for origins, so m_f still averages over a sizeable stretch of the flight rather than a handful of displacements. Left: the equal-flight TA-MSD of all available flights. Trajectories sit on the common 10-second grid, so origins shift in 10-second steps; the plotted lags are a log-spaced subset. Shading shows pointwise 90 percent bootstrap intervals. Right: the number of contributing flights relative to the count at ten seconds.

Up to a few thousand seconds almost every flight contributes; from about 6,000 seconds the count falls steeply. Paragliders go from 155,085 flights at ten seconds to 46,273 at 10,000 seconds and 27 at 32,690 seconds; hang gliders from 6,060 to 2,326 and then 20 at 24,740 seconds. The dotted line marks a tenfold loss. The curves end where fewer than 20 site-day groups still contribute. A group is the set of flights from one take-off site on one calendar day; the bootstrap resamples whole groups, so with fewer than 20 the confidence bands would not be meaningful.

The circled end of the MSD is where this loss acts. There the curve steepens: around 6,600 seconds the local slope of the all-flight curve gives H near 0.90 for paragliders, whereas the fixed cohort of the next slide gives about 0.70 at the same lag. Same estimator, different flights: the steepening comes from which flights are averaged, not only from how they move. An H fitted to the all-flight curve therefore

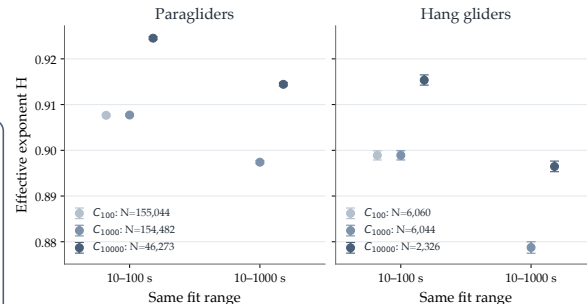
C_{10000} : select once for the largest lag

$\tau_{\max} = 10,000 \text{ s} \leq 0.8 T_s$: $T_s \geq 12,500 \text{ s}$
Only these segments enter, at every lag.

Advantages of C_{10000}

- 1 Same flights and segments, 10 to 10,000 s, with no loss of flights beyond $6 \times 10^3 \text{ s}$
- 2 Three decades of lag; at $\tau_{\max}$ each segment still offers at least 251 origins for m_f
- 3 Sample still large, same order as at 10 s:
46,273 paragliders (30%)
2,326 hang gliders (38%)

Cost of selecting longer flights



Nested cohorts, identical fit ranges, paired bootstrap draws.

Long flights raise H by 0.016–0.018 on matched lag ranges; origins still decrease with lag.

About 2% in H buys the same flights at every lag over three decades.

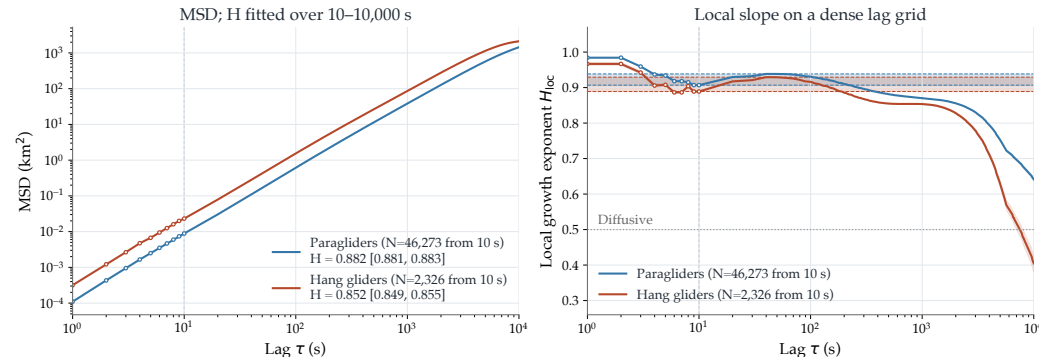
Appendix: can we afford a fixed population?

C_{10000} answers the three desiderata of the previous slide. We apply the admission rule once, at the largest lag: $10,000 \text{ s} \leq 0.8 T_s$ means segments of at least 12,500 seconds on the common grid. Even at the maximum lag, each segment keeps 2,500 seconds of origins, at least 251 on the 10-second grid. Only these segments enter, at every lag; shorter segments of an otherwise eligible flight are excluded throughout. Between 6,000 and 10,000 seconds, where the all-flight curve loses most of its flights, this cohort loses none. The cohort is exactly the set of flights still present at 10,000 seconds on the previous slide: 46,273 paragliders and 2,326 hang gliders, 30 and 38 percent of the flights available at ten seconds. So the sample stays of the same order as the full population, while covering three decades of lag. Pushing the maximum lag further would cost flights quickly: beyond about 16,000 seconds fewer than a tenth remain.

The price is a duration selection: long flights are not a random subset. The figure measures it by comparing nested cohorts over identical fit intervals, using paired bootstrap draws. The main cohort increases H by 0.0164–0.0177, about 1.8–2.0 percent. We accept this small, measured shift to keep the population constant across three decades of lag. Origins per flight still vary with lag. Source: thesis Section 3.1.3 and ch3_transport_report.json.

C_{10000} : changing support below 10 s; the same flights at every lag from 10 to 10,000 s.

$M_2(\tau) \propto \tau^{2H}$: $H = \frac{1}{2}$ diffusion $H = 1$ ballistic motion



$H_{\text{PG}} = 0.882$ [0.881, 0.883] > $H_{\text{HG}} = 0.852$ [0.849, 0.855] over 10–10,000 s

Nominal 90% bootstrap CIs; clusters sampled with replacement. Clustering remains too fine; bands likely too narrow.

Displacement growth from 1 s to 10,000 s

The parent population is the original C10000 cohort, with the same selected long segments. Below ten seconds, a segment contributes when its native logging interval is no larger than the lag. Thus the contributing population grows from 33,450 paragliders and 690 hang gliders at one second to the full 46,273 and 2,326 at ten seconds. It stays fixed thereafter, through 10,000 seconds.

At short lags we use native origins and linearly interpolate endpoints when the lag is not a multiple of the logging interval, always within the selected segment span. This monotone admission rule differs from Figure 3.1: that figure uses the general population and requires exact native short-lag resolution, so its short-lag counts need not grow monotonically. From ten seconds, this slide uses the thesis's common ten-second grid and reproduces every original MSD point.

There are 277 evaluated lags, including every second from one to ten. Local H is half the log-log slope in a plus/minus 0.25-decade window, expanded to the nearest three lags when sparse. The one-second estimate uses one to three seconds. Slopes below ten seconds, and windows crossing that boundary, combine dynamics with changing sample composition. Lines connect computed estimates. In the right panel, each shaded horizontal band spans that discipline's local-slope range from its local minimum near ten seconds to its local maximum near 40–50 seconds.

The original 10–10,000 second fits and 1,000 paired site-day bootstrap draws are preserved: $H=0.882$ [0.881, 0.883] and 0.852 [0.849, 0.855]. Bands are pointwise 90 percent intervals. Sources: thesis Section

Appendix: bootstrap confidence intervals

1 Group flights

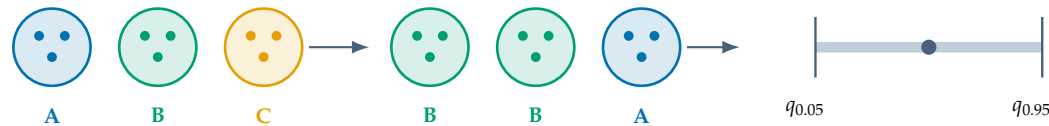
Take-off site $\times$ calendar day.
Aim: independent groups.

2 Resample groups

Sample with replacement.
Keep each group together.

3 Recompute

1,000 replicates.
Take the 5th–95th percentiles.



Current clusters are too small to absorb all shared flight conditions.

Intervals likely underestimate uncertainty; clustering is still being refined

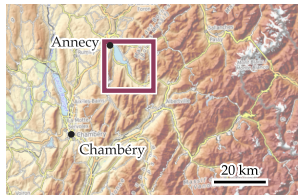
Schematic resample. Intervals are pointwise; they do not guarantee simultaneous curve coverage.

The reported confidence intervals use the current site–day cluster bootstrap. Each group combines flights from the same declared take-off site and calendar day. We sample groups with replacement, preserve all flights belonging to a sampled group, recompute the equal-flight estimator, and repeat this one thousand times. The same multiplicities are paired across lags and conditional comparisons. The 5th and 95th percentiles form nominal 90 percent intervals; for a curve they are pointwise, not a simultaneous confidence band.

Clustering aims to keep dependent flights together and make different groups approximately independent. The current partition is too fine: nearby sites and neighbouring days can share weather and other conditions, so flights in different groups can still be dependent. In the current cohort, 71.1 percent of paraglider groups and 81.3 percent of hang-glider groups are singletons. The bands are consequently likely to understate uncertainty; their nominal level should not be read as demonstrated coverage.

Validation with 50-km cells crossed with initial-altitude class and month of year gives standard errors for fitted H roughly four times the site–day value for paragliders and twice for hang gliders. Further aggregation changes the errors again; independence and uncertainty have not stabilised. These are diagnostic comparisons, not a validated replacement partition, and other chapter bands have not been recomputed with that scheme. All intervals in this presentation retain the original bootstrap convention. Sources: thesis Section 3.1.2; ch3_transport_grid_bootstrap.json and its generated values.

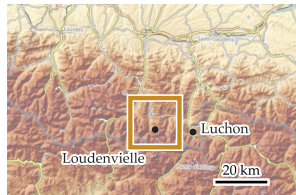
Alps: Lake Annecy



Centre (WGS84)
 45.82374° N
 6.22085° E

Bounds (Lambert-93, m)
E 940000–960000
N 6520000–6540000

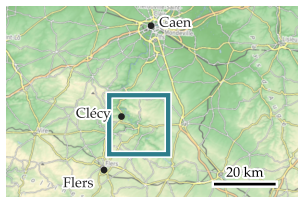
Pyrenees: Louron Valley



Centre (WGS84)
 42.77884° N
 0.43624° E

Bounds (Lambert-93, m)
E 480000–500000
N 6180000–6200000

Channel Coast: Suisse Normande



Centre (WGS84)
 48.88143° N
 0.40990° W

Bounds (Lambert-93, m)
E 440000–460000
N 6860000–6880000

Champagne-Lorraine: Mouzon



Centre (WGS84)
 49.63141° N
 5.07544° E

Bounds (Lambert-93, m)
E 840000–860000
N 6940000–6960000

Appendix: locations of the 20 km cells

These are the exact four cells shown on slide 11, each selected as the busiest 20 by 20 km block in its regional box. Their coloured outlines show the complete cell, with surrounding terrain and named reference communes for orientation. The Alpine cell covers Lake Annecy in the northern French Alps. The Pyrenean cell covers the Louron Valley around Loudenvielle. The Channel Coast cell lies inland around Clécy in Suisse Normande. The Champagne-Lorraine cell lies around Mouzon in the Ardennes.

The Easting and Northing intervals printed on the slide are the exact rectangle bounds in metres in Lambert-93, EPSG:2154. The geographic coordinates are the transformed cell centres in WGS84, shown to five decimal places. They are not corner coordinates: a Lambert-93 square is not an axis-aligned latitude–longitude box. Full-precision centres and all four transformed corners are saved in cell-locator-report.json. Map up follows Lambert-93 grid north; each map includes a 20 km scale bar. The two mountain views span 120 by 78 km; the two lowland views span 100 by 65 km.

Sources: plane-cells-report.json for the cell definitions; archived OpenTopoMap basemaps and their retrieval metadata in assets/regional-climb; French government geographic API for the labelled municipal centres. Locality context: Saint-Lary Tourism and Suisse Normande, Clécy. The map crops, outlines and coordinates are reproduced offline by render_cell_locators.py.

Flight archive

FFVL Coupe Fédérale de Distance: IGC tracks, dates, declared sites, circuits and wing classes.
parapente.ffvl.fr/cfd/liste delta.ffvl.fr/cfd/liste

Maps and terrain

OpenTopoMap: regional relief and contours (OSM / SRTM, CC-BY-SA).
IGN / Géoplateforme: local Plan IGN backgrounds.
IGN RGE ALTI: local terrain and derived crests.

Locator and analysis

Natural Earth: France outline, public domain.
MSD, intervals and crossings: this thesis's prepared pipeline.
Code, figure scripts and recorded source provenance

Data and geographic sources

The FFVL archive supplies the observed trajectories and catalogue metadata. Paraglider and hang-glider collections have separate portals and season ranges, shown earlier. The pipeline applies the documented cleaning, selection and equal-flight estimators; generated figures are outputs of this thesis rather than independent external datasets. Thermal crossings in the five-kilometre cells come from the prepared viewer store. Regional climb-use maps come from the C10000 paraglider cohort and archived HMM phase fixes.

Regional maps use OpenTopoMap tiles, including OpenStreetMap map data and SRTM relief, warped to Lambert-93; the tiles and style are credited under CC-BY-SA. The background is desaturated and lightened for display. IGN Plan IGN is the coloured background of the separate small-cell maps. Approximate local crest lines are computed independently of flight positions from RGE ALTI terrain, requested at 25 m sampling for twelve selected five-kilometre cells with a 500 m processing buffer. The France locator uses Natural Earth country boundaries in the public domain.

Named summit points from IGN BD TOPO were tested earlier but are not the crest overlay currently displayed. The thermal-mechanism image is a schematic, not an atmospheric measurement. Source hashes, extraction parameters and figure scripts are recorded with the presentation and thesis.